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1.0 EVOLUTION OF OPERATING SYSTEMS

1.0.1 Introduction

Operating systems (OS) are the backbone of computer systems, managing hardware resources and providing services for application software. The evolution of OS reflects the progress in computing technology, from large mainframes to modern mobile and cloud-based systems.

1.1 HISTORICAL DEVELOPMENT

1950s – Early Computers and Batch Processing

Early computers had no operating system; programs were loaded manually.

Batch processing systems were developed to run jobs sequentially, reducing idle time.

Key milestones: IBM 701 (1952), first mainframes.

1960s – Multiprogramming and Time-Sharing

OS began supporting multiprogramming: multiple programs in memory simultaneously.

Time-sharing systems allowed multiple users to interact with the computer at the same time.

Example: CTSS (Compatible Time-Sharing System) at MIT.

1970s – Unix and Minicomputers

Unix introduced by Ken Thompson and Dennis Ritchie at Bell Labs (1969).

Unix emphasized portability, multi-user support, and hierarchical file systems.

Minicomputers became affordable, spreading OS concepts beyond mainframes.

1980s – Personal Computers and GUI

Rise of PCs required OS for individual users.

Examples:

MS-DOS (Microsoft) – command-line interface Mac

OS (Apple) – graphical user interface (GUI)

Networking features and multitasking started to appear.

1990s – Networked OS and Linux

OS became network-aware, supporting Internet and local networks.

Linux by Linus Torvalds (1991) introduced an open-source, Unix-like OS.

Windows evolved: Windows 95 brought GUI, plug-and-play hardware support, and multitasking.

2000s – Mobile and Embedded OS

Rise of smartphones required lightweight OS.

Examples: Symbian, Blackberry OS, iOS (2007), Android (2008)

OS optimized for power efficiency, touch interfaces, and app ecosystems.

2010s–Present – Cloud, Virtualization, and Security

Cloud-based OS and virtual OS environments became common (VMware, Hyper-V).

Mobile OS dominance: Android and iOS

Focus on security, user experience, and cross-device integration.

2.0 Real-Time Operating Systems

2.0.1 Introduction

A Real-Time Operating System (RTOS) is an operating system designed to process data and respond to events within a strict time limit. In a real-time system, correct output is not enough. The output must also be produced within a defined deadline. If the deadline is missed, the system is considered to have failed.

Real-time systems are mainly used in environments where timing is critical, such as industrial machines, robotics, medical devices, and automotive control systems.

2.1 REQUIREMENTS OF A REAL-TIME OPERATING SYSTEM

A real-time operating system must meet the following requirements:

- Predictable Response Time
  The system must respond to events within a guaranteed time frame.
- Deterministic Behavior
  The execution time of tasks should be known and consistent.
- Priority-Based Scheduling
  Higher priority tasks must execute before lower priority tasks.
- Minimal Interrupt Latency
  The delay between an interrupt and its handling must be very small.
- Reliability and Stability
  The system must operate continuously without crashing.

2.2 TYPES OF REAL-TIME SYSTEMS

2.2.1 Hard Real-Time Systems

Missing a deadline can cause complete system failure.

Example: Airbag control system in a car.

2.2.2 Soft Real-Time Systems

Missing a deadline reduces performance but does not cause total failure.
Example: Video streaming systems.

3.0 MOBILE OPERATING SYSTEMS

A Mobile Operating System is designed for smartphones and tablets. It manages hardware resources and provides a platform for mobile applications. The most common mobile operating systems are **Android** and **iOS**.

3.1 DESIGN CONSIDERATIONS

- Mobile OS must handle:
- Low battery consumption
- Limited RAM and CPU
- Touch screen interface
- Wireless connectivity
- App sandboxing for security

3.2 UNIQUE FEATURES

- App store support
- Push notifications
- Biometric authentication
- Sensor integration

3.3 SECURITY CONCERNS

- Malware attacks
- Data privacy risks
- OS fragmentation
- Network security threats

4.0 PRACTICAL TASK: THREAD ANALYSIS OF RUNNING APPLICATIONS

To understand how an operating system manages processes and threads, three different applications were executed on the system. For each application, thread-related information was observed using Task Manager and Resource Monitor.

4.1 APPLICATION 1: WEB BROWSER (CHROME)

Application Name: Google Chrome

Total Threads Observed: 22

Sample Thread IDs: 30072

CPU Usage of Threads: 0.00

System Interrupts	-	Deferred Procedure Calls and Interrupt Service Routines	Running	-	0	1.99
System	4	NT Kernel & System	Running	354	1	1.70
dwm.exe	1752	Desktop Window Manager	Running	21	1	1.56
procexp64.exe	18680	Sysinternals Process Explorer	Running	9	0	1.36
MsMpEng.exe	4740		Running	58	0	1.26
chrome.exe	28296	Google Chrome	Running	48	0	0.59
sqlservr.exe	2372	SQL Server Windows NT - 64 Bit	Running	93	0	0.50
svchost.exe (NetworkService -p)	2080	Host Process for Windows Services	Running	13	0	0.49
sqlservr.exe	14068	SQL Server Windows NT - 64 Bit	Running	93	0	0.45
audiodg.exe	8396	Windows Audio Device Graph Isolation	Running	20	0	0.42
chrome.exe	30072	Google Chrome	Running	22	0	0.42
explorer.exe	9304	Windows Explorer	Running	147	0	0.40
svchost.exe (osprivacy -p)	6908	Host Process for Windows Services	Running	9	0	0.37
svchost.exe (LocalServiceNoNet...	4444	Host Process for Windows Services	Running	26	0	0.26
xampp-control.exe	876	xampp-control	Running	2	0	0.26
svchost.exe (netprofm -p)	2556	Host Process for Windows Services	Running	73	0	0.21

Figure 1.0 : Google Chrome

4.2 APPLICATION 2: VS CODE

Application Name: VS Code

Number of Threads: 23

Thread IDs: 5472

CPU Usage per Thread: 0.00

smartscreen.exe	29720	Windows Defender SmartScreen	Running	6	0	0.00
Code.exe	27796	Visual Studio Code	Running	7	0	0.00
Code.exe	28092	Visual Studio Code	Running	20	0	0.00
Code.exe	5472	Visual Studio Code	Running	23	0	0.00
Code.exe	5196	Visual Studio Code	Running	26	0	0.00
Code.exe	3120	Visual Studio Code	Running	28	0	0.00
svchost.exe (WerSvcGroup)	27692	Host Process for Windows Services	Running	7	0	0.00
msedgewebview2.exe	6968	Microsoft Edge WebView2	Terminated	16	0	0.00

Figure 2.0 : Vs Code

4.3 APPLICATION 3: QUICK SHARE

Application Name: Quick Share

Number of Threads: 197

Thread IDs: 19668

CPU Usage per Thread: 197

MicrosoftLightbox.exe	13612	MicrosoftLightbox	Suspended	11	0	0.00
AccountsControlHost.exe	22752	Accounts Control Host	Suspended	16	0	0.00
LockApp.exe	26648	LockApp.exe	Suspended	17	0	0.00
backgroundTaskHost.exe	22316	Background Task Host	Suspended	12	0	0.00
SystemSettings.exe	29296	Settings	Suspended	45	0	0.00
nearby_share.exe	19668	Quick Share	Running	197	8	6.44
System Interrupts	-	Deferred Procedure Calls and Interrupt Service Routines	Running	-	1	1.70
perfmon.exe	12748	Resource and Performance Monitor	Running	17	1	1.64
System	4	NT Kernel & System	Running	353	1	1.30
dwm.exe	1752	Desktop Window Manager	Running	19	1	0.89
sqlservr.exe	2372	SQL Server Windows NT - 64 Bit	Running	98	0	0.58
sqlservr.exe	14068	SQL Server Windows NT - 64 Bit	Running	100	0	0.50
audiodg.exe	8396	Windows Audio Device Graph Isolation	Running	31	0	0.42

Figure 3.0 : Quick Share